

Subject Description Form

Subject Code	LSGI2801
Subject Title	Smart Cities: Context, Policy and Government
Credit Value	3
Level	2
Pre-requisite/ Co-requisite/ Exclusion	Nil
Objectives	This course aims to provide students with a foundational understanding of the core concepts of smart cities, the key technologies that underpin their development, and government structures, policy contexts, and notions of citizenship in shaping smart city initiatives. The course will also provide an overview of various smart city solutions from the six main areas of the Hong Kong Smart City Blueprint, namely, Smart Mobility, Smart Living, Smart Environment, Smart People, Smart Governance, and Smart Economy.
Intended Learning Outcomes	Upon completion of the subject, students will be able to: a. Identify salient spatial, technological and governance characteristics of smart cities b. Obtain in-depth knowledge of emerging technologies and solutions for smart city developments c. Participate in a social science discourse on smart city developments in Hong Kong and beyond d. Evaluate smart city initiatives and solutions with respect to contents, sectors, governance and citizenship e. Develop ideas to design smart city initiatives
Subject Synopsis/ Indicative Syllabus	1. Introduction to smart cities • Smart city concepts, characteristics, and components • History and evolution of smart cities 2. Smart city infrastructures and technologies • Urban sensing and internet of things • Spatial big data analytics • Artificial intelligence • Information and communication technologies (ICTs) • 3D spatial data infrastructure

	• Cloud & edge computing • Urban computing • Robotics 3. Smart city solutions – perspectives from governments, industries, and academia. • Smart government • Smart environment • Smart mobility • Smart economy • Smart living • Smart people 4. Student innovation on future smart cities • Direct students to propose a new idea on smart cities in one of the six areas mentioned above (or beyond) using the latest technologies • Write a report on the feasibility, design and implementation of the idea • Present on the design to the whole class and invite guests from industry or public sectors who are interested in smart cities proposals
Teaching/Learning Methodology	1. Lectures and seminars to explain smart city concepts, government practices, and technological innovations in smart cities. 2. Extracurricular reading and in-class presentations to reinforce learning experiences 3. A final report that allows students to provide a comprehensive review of smart city development & initiatives in contemporary cities, along with their proposed urban solutions by incorporating the latest smart city technologies.

Assessment Methods in Alignment with Intended Learning Outcomes	Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed (Please tick as appropriate)				
			a	b	c	d	e
	1. Assignment	30%	✓	✓	✓	✓	✓
	2. Final project	70%	✓	✓	✓	✓	✓
	Total	100 %					
	Explanation of the appropriateness of the assessment methods in assessing the intended learning outcomes: The continuous assessment includes: (1) an assignment that assess students’ understanding of smart city concepts, government practices, and technological innovations, and (2) a final project report that provides a comprehensive review of smart city development & initiatives in contemporary cities, along with their proposed urban solutions by incorporating the latest smart city technologies.						
Student Study Effort Expected	Class contact:						
	▪ Lectures					26 Hrs.	
	▪ Tutorials					13 Hrs.	
	Other student study effort:						
	▪ Reading of reference materials					40 Hrs.	
	▪ Writing project report					40 Hrs.	
	Total student study effort					106 Hrs.	
Reading List and References	• Noveck, Beth Simone. Smart citizens, smarter state: The technologies of expertise and the future of governing. Harvard University Press, 2015. • Batty, Michael, et al. "Smart cities of the future." The European Physical Journal Special Topics 214.1 (2012): 481-518. • Wenzhong Shi, et al. “Urban Informatics” (https://link.springer.com/book/10.1007/978-981-15-8983-6) • Hong Kong Smart City Blueprint (https://www.smartcity.gov.hk/)						

	• Nam, Taewoo, and Theresa A. Pardo. "Conceptualizing smart city with dimensions of technology, people, and institutions." Proceedings of the 12th annual international digital government research conference: digital government innovation in challenging times. ACM, 2011. • Townsend, Anthony M. Smart cities: Big data, civic hackers, and the quest for a new utopia. WW Norton & Company, 2013. • Batty, Michael. Inventing Future Cities. MIT Press, 2018.
--	---